The two noncommutative Q-Gauss summations

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Abstract

We justify the limit in Schlosser's terminating noncommutative Q-Pfaff–Saalschütz identities and prove both parts of Conjecture 7.1. A common geometric majorant works for the two orders of multiplication. A block-matrix argument includes noninvertible bases Q.

Task 1: statement in explicit product notation

Let R be a unital Banach algebra. Let A, B, C be invertible, and suppose that Q commutes with A, B, C, and that $D = BC^{-1}A$ also commutes with A, B, C. Put

$$Z = A^{-1}CB^{-1} = D^{-1}, \qquad q = \|Q\| < 1, \quad z = \|Z\| < 1.$$

As in the source, all displayed denominator factors are assumed invertible; here this requires $I - CQ^j$ invertible for every $j \geq 0$. The factors $I - Q^{j+1}$ and $I - ZQ^j$ are automatically invertible. The condition $\|Q\| < 1$ is the standing convention immediately before Conjecture 7.1 in Section 7 of [1]; it is not an added restriction. The norm may be taken to satisfy $\|I\| = 1$: if needed, the equivalent left-regular operator norm has this property and does not increase $\|Q\|$ or $\|Z\|$.

The element Z commutes with A, B, C, Q. Consequently

$$C = ABZ, \qquad CB^{-1} = AZ, \qquad A^{-1}C = BZ.$$

For $j \geq 0$, define

$$\begin{aligned} X_j &= (I - CQ^j)^{-1}(I - AQ^j)(I - BQ^j), \\ Y_j &= (I - CQ^j)^{-1}(I - AZQ^j)(I - ZQ^j)^{-1}(I - BZQ^j), \\ V_j &= (I - CQ^j)^{-1}(I - AZQ^{j+1})(I - ZQ^j)^{-1}(I - BZQ^{j+1}), \\ H &= (I - AZ)(I - BZ), \qquad K_k = \prod_{j=1}^{k}(I - Q^j)^{-1}. \end{aligned}$$

All products written with increasing indices are read from left to right. Set $T_0^{\mathrm{I}} = T_0^{\mathrm{II}} = I$, and, for $k \geq 1$, set

$$T_k^{\mathrm{I}} = X_{k-1} \cdots X_0 K_k, \qquad T_k^{\mathrm{II}} = X_0 \cdots X_{k-1} K_k.$$

These are exactly the coefficients of the source's two ${}_2\phi_1$ series after their central argument Z has been moved to the right.

Theorem 1. *Under these hypotheses both series and both products below converge in norm, and*

$$\sum_{k=0}^{\infty} T_k^{\mathrm{I}} Z^k = \lim_{n\to\infty} Y_{n-1}\cdots Y_0, \tag{1}$$

$$\sum_{k=0}^{\infty} T_k^{\mathrm{II}} Z^k = \Big(\lim_{n\to\infty} V_0\cdots V_{n-1}\Big) H. \tag{2}$$

Thus both identities (7.1) and (7.2) of Schlosser's Conjecture 7.1, equivalently MathDB 378712, hold. In particular, Q need not be invertible.

The terminating identities with nonnegative powers

For $0 \le k \le n$, put

$$W_{n,k} = Z^k \prod_{r=n-k}^{n-1} (I - Q^{r+1})(I - ZQ^r)^{-1}. \tag{3}$$

The empty product is I. All the factors in $W_{n,k}$ commute with all the X_j and with one another.

Lemma 1. *For every sufficiently large integer n,*

$$\sum_{k=0}^{n} T_k^{\mathrm{I}} W_{n,k} = Y_{n-1}\cdots Y_0, \tag{4}$$

$$\sum_{k=0}^{n} T_k^{\mathrm{II}} W_{n,k} = V_0\cdots V_{n-1}(I - BZQ^n)^{-1}(I - AZQ^n)^{-1}H. \tag{5}$$

Proof. First suppose Q is invertible. Apply Theorem 5.2 of [1], with upper parameters A, B, Q^{-n}, lower parameters $C, Z^{-1}Q^{1-n}$, and argument Q. Its sum terminates at $k = n$. In each summand, the additional factors, at exponent j with $0 \le j < k$, can be moved past the A, B, C factors and rewritten as

$$(I - Z^{-1}Q^{1-n+j})^{-1}(I - Q^{j-n})Q = Z(I - Q^{n-j})(I - ZQ^{n-j-1})^{-1}. \tag{6}$$

For example, factor the denominator on the left as $-Z^{-1}Q^{1-n+j}(I - ZQ^{n-j-1})$ and the numerator as $-Q^{j-n}(I - Q^{n-j})$. Their negative powers cancel exactly. Multiplication over $j = 0, \ldots, k-1$ gives (3). The two right sides of Theorem 5.2 are exactly the right sides of (4) and (5), respectively. All denominators in the terminating sums exist, by (6) and $z, q < 1$. Choose n large enough that $\|AZ\|q^n < 1$ and $\|BZ\|q^n < 1$; the two extra inverses in (5) then also exist.

To remove invertibility of Q, fix such an n and work in $M_2(R)$, with the maximum row-sum norm. Replace each $U \in \{A, B, C, Z\}$ by $\widehat{U} = \mathrm{diag}(U, U)$ and replace Q by

$$Q_\varepsilon = \begin{pmatrix} Q & \varepsilon I \\ \varepsilon I & 0 \end{pmatrix}, \qquad Q_\varepsilon^{-1} = \begin{pmatrix} 0 & \varepsilon^{-1} I \\ \varepsilon^{-1} I & -\varepsilon^{-2} Q \end{pmatrix} \quad (\varepsilon \neq 0).$$

These replacements preserve every commutation relation and $\widehat{C} = \widehat{A}\widehat{B}\widehat{Z}$. Moreover $\|Q_\varepsilon\| \le q + |\varepsilon| < 1$ for sufficiently small ε. At $\varepsilon = 0$, the finitely many factors $I - \widehat{C}Q_0^j$ needed for the two identities are invertible: for $j = 0$ both diagonal blocks are $I - C$, and for $j \ge 1$ the blocks are $I - CQ^j$ and I. The terminal factors have the same property. Invertibility is open, so these finitely many inverses persist for sufficiently small ε. Apply the invertible-base identities just established, written entirely in nonnegative powers, and let $\varepsilon \to 0$. Each side is a continuous finite expression. Its upper-left diagonal block at the limit is the corresponding expression for the original Q. This proves both identities for arbitrary Q. No density assertion about invertibles in R is used. □

A uniform bound and the limit

For large j we have $\|C\|q^j \le 1/2$ and therefore

$$\|X_j - I\| = \|(I - CQ^j)^{-1}\big((C - A - B)Q^j + ABQ^{2j}\big)\| \le 2\big((\|C\| + \|A\| + \|B\|)q^j + \|A\|\|B\|q^{2j}\big).$$

Thus $\sum_{j\ge 0}\|X_j - I\| < \infty$. In particular, the number

$$L = \prod_{j=0}^{\infty}(1 + \|X_j - I\|)\prod_{j=1}^{\infty}(1 - q^j)^{-1}$$

is finite and bounds $\|T_k^{\mathrm{I}}\|$ and $\|T_k^{\mathrm{II}}\|$ for every k. This already proves absolute convergence of both series in the theorem.

Neumann's bound applied to (3) gives, uniformly in $0 \le k \le n$,

$$\|W_{n,k}\| \le z^k \prod_{r=n-k}^{n-1}\frac{1 + q^{r+1}}{1 - zq^r} \le Mz^k, \qquad M = \prod_{r=0}^{\infty}\frac{1 + q^{r+1}}{1 - zq^r} < \infty. \tag{7}$$

The last scalar product is finite because its factors are positive, their deviations from 1 are summable, and $z < 1$. For each fixed k, every factor of the finite product in (3) tends to I as $n \to \infty$, so $W_{n,k} \to Z^k$. Extend $T_k^\sigma W_{n,k}$ by zero for $k > n$, where $\sigma \in \{\mathrm{I}, \mathrm{II}\}$. The common summable bound LMz^k proves

$$\lim_{n\to\infty}\sum_{k=0}^{n} T_k^\sigma W_{n,k} = \sum_{k=0}^{\infty} T_k^\sigma Z^k. \tag{8}$$

For completeness, this follows directly by choosing a common tail $\sum_{k>K} LMz^k$ small and then passing to the limit in the finitely many terms $k \le K$; no commutative summation theorem is needed.

Similarly $Y_j - I = O(q^j)$ and $V_j - I = O(q^j)$ in norm, since the inverses in their definitions are $I + O(q^j)$ for large j. If $\sum_j\|E_j - I\| < \infty$, the finite products of E_j are bounded by $\prod_j(1 + \|E_j - I\|)$, and a telescoping estimate makes both the ascending and descending products Cauchy. Consequently both product limits in the theorem exist. Finally

$$(I - BZQ^n)^{-1}(I - AZQ^n)^{-1}H \longrightarrow H.$$

Pass to the limit in Lemma 1, using (8) and the bounded product prefixes. This proves (1) and (2). □

Tasks 2–5: proposed follow-up problems

The following are proposed further tasks, not additional solved results.

Task 2. Determine whether both identities remain valid when $\|Q\| < 1$ and $\|Z\| < 1$ are replaced by $r(Q) < 1$ and $r(Z) < 1$, where r denotes spectral radius, with the same inverse hypotheses.

Task 3. Derive an explicit computable bound for the error between each infinite sum and its nth terminating approximation in Lemma 1, in terms of q, z, A, B, C and a finite list of resolvent norms.

Task 4. Retain the stated commutation relations and $C = ABZ$, but dispense with invertibility of the five parameters. Prove the two finite identities by induction whenever their denominator factors are invertible.

Task 5. Classify the boundary cases with central Z, $r(Z) = 1$ and $r(Q) < 1$ for which the type I or type II series converges, and determine whether a corresponding product or Abel-limit formula holds.

References

[1] M. J. Schlosser, *Summation formulas for noncommutative hypergeometric series*, in Z. Nashed and X. Li (eds.), *Frontiers in Orthogonal Polynomials and q-Series*, World Scientific, 2018, pp. 451–496, doi:10.1142/9789813228887_0024. Author's updated 36-page version: author-hosted PDF. Original 35-page version: `https://arxiv.org/abs/math/0411136`. Theorem 5.2 and Conjecture 7.1 have the same numbering in both.